MIL-C-50652/2A (AR)
09 April 1993
SUPERSEDING
MIL-C- 50652 /2(MU)
15 April 1971

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 10548007

(GENERATOR, COMPUTER TIMING)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** The detail requirements of this specification sheet together with the general requirements of MIL-C-50652 comprise the complete requirements for the Computer Timing Generator. This assembly uses binary counters in cascade to produce timing pulses of 4 KHz, 250 HZ, and 25 Hz. Also provided by this assembly is a signal conditioner and a five-line decoder output.

2* APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-50652	**Circuit Card Assemblies, General Specifications for**

Beneficial comments (recommendations, additions, deletions and any pertinent information that may be of use in improving

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Us. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548007 - Computer Timing Generator

SPECIAL PACKAGING INSTRUCTION

PS 10548007 - packaging of Computer Timing Generator

(Copies of other Government documents, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.2.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548007, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Environmental service conditions. As specified in MIL-C-50652.

3.4. Electrical. Each conductor shall be electrically

assembly.

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3.5 Performance. When supplied with the input power and signals given Table I the computer timing generator shall yield the output signals as specified in Table II and III. With Manual Target Reject or R at logic O, the 200M and ZOOM outputs shall be logic 1 and logic O respectively. With Target Data at logic O, the TG SET output shall be logic 1 and the TG RESET output shall be pulses with the same width and spacing as those on T1/250. Signals specified as logic levels or composed of logic level shall have a logic I level +3.5 ± 1.0 volts (V) and for a logic O level O + 0.5 V.

3.6 First Article. As specified in MIL-C-50652.

3.7 Soldering. As specified in MIL-C-50652.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 apply to the assembly covered by this detail specification sheet of the circuit card assembly.

4.2 First article. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

4.5 Inspection equipment. As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.3 Test equipment. In conjunction with 4.5.1 and 4.5.2 the test equipment listed in Table IV shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE I. - INPUT SIGNAL CHARACTERISTICS.

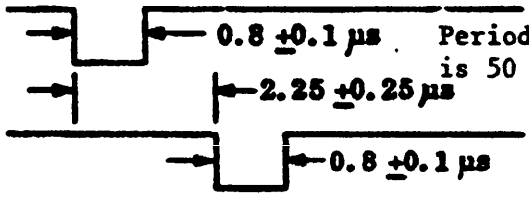
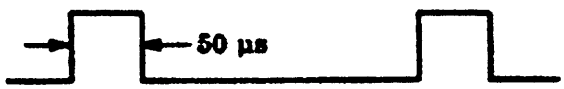
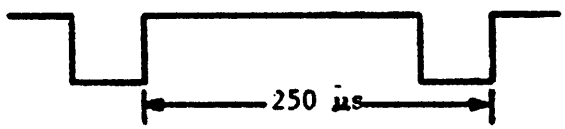
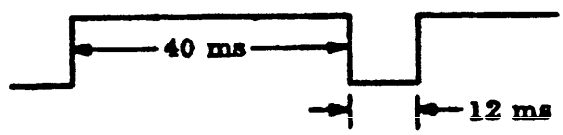
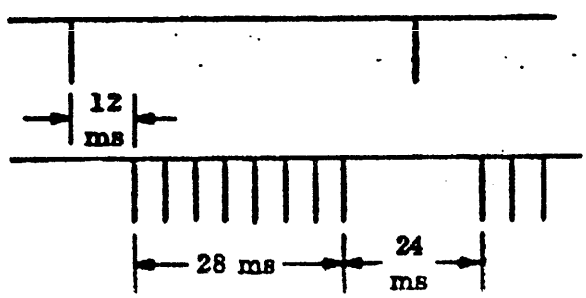
SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
Input power	5.0 ±0.3 Vdc, regulated; ac ripple, less than 50 mV peak-to-peak; capacity 300 mA.	P1-30, P1-31, & P1-32 (+5V) P1-24, P1-25, & P1-26 (+5V RET)
Input signals		
$\overline{T1}$		P1-22
$\overline{T3}$	0.8 ±0.1 µs	P1-15
TARGET DATA	Logic 1	P1-9
$\overline{R200/T2}$	Connected to $\overline{TG\ SET}$ (P1-7)	P1-14
MANUAL TARGET REJECT	LOGIC 1	P1-20
$\overline{R\ MAX}$	LOGIC 1	P1-34

TABLE II. - OUTPUT SIGNAL CHARACTERISTICS

SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
Output signals		
T1	Complement of $\overline{T1}$ 	P1-10
T3	Complement of $\overline{T3}$ 	P1-16
T3/4KC	Logic 1 pulse being coincident with T3 and having the same characteristics but occurring once for every five T3 pulses.	P1-19
T1/250	Logic 0 pulse being coincident with T1 and having the same characteristics but occurring every 4.0 + 0.5 msec	P1-18

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TABLE III. - OUTPUT SIGNAL WAVEFORMS.

SIGNAL NAME	WAVESHAPE OR CHARACTERISTIC	TERMINALS
	For F4/5 through F1/5 each division is equal to 50 µs.	
F4/5		P1-45
$\overline{F4/5}$		P1-47
F2/5		P1-4
$\overline{F2/5}$		P1-2
F1/5		P1-13
$\overline{F1/5}$		P1-11
200M		P1-50
$\overline{200M}$		P1-55
$\overline{TG\ SET}$		P1-7
$\overline{TG\ RESET}$		P1-5

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TABLE IV. TEST EQUIPMENT LIST.

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ± 0.3 Vdc, regulated. Ac ripple: less than 50 mv peak-to-peak Capacity: 300 mA.
2. Pulse generator	Frequency: 20 kHz minimum Duty factor: 10% minimum Amplitude: Minimum of 4 V into a SO-ohm load. pulse rise and fall times less than 0.1 ms.
3. Digital voltmeter	With accuracy and resolution of ± 0.2% or better from 0.0090 to 99.0 Vdc. Input impedance 10 megohms or greater on all voltage scales.
40 Test fixture	See Figure 1. See Table V for test fixture parts list.
5. Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less. sweep period 50 ns per division or less, input resistance greater than 8 megohms, input capacitance less than 25 PF.

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TABLE V. TEST FIXTURE PARTS LIST,

DESIGNATION	NOMENCLATURE AND DESCRIPTION
El	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent.
	(1) 5893232-1, Malco Mfg. CO., Inc. Chicago, Illinois.
	(2) 500946-01, National Connector Corp. Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, PA
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
Z1	Integrated circuit, Hex Inverter SN5404 or equivalent.
S1 & S2	Switch SPDT.
TPI thru TP16	Tip jack, H. H. Smith 1501 or equivalent.

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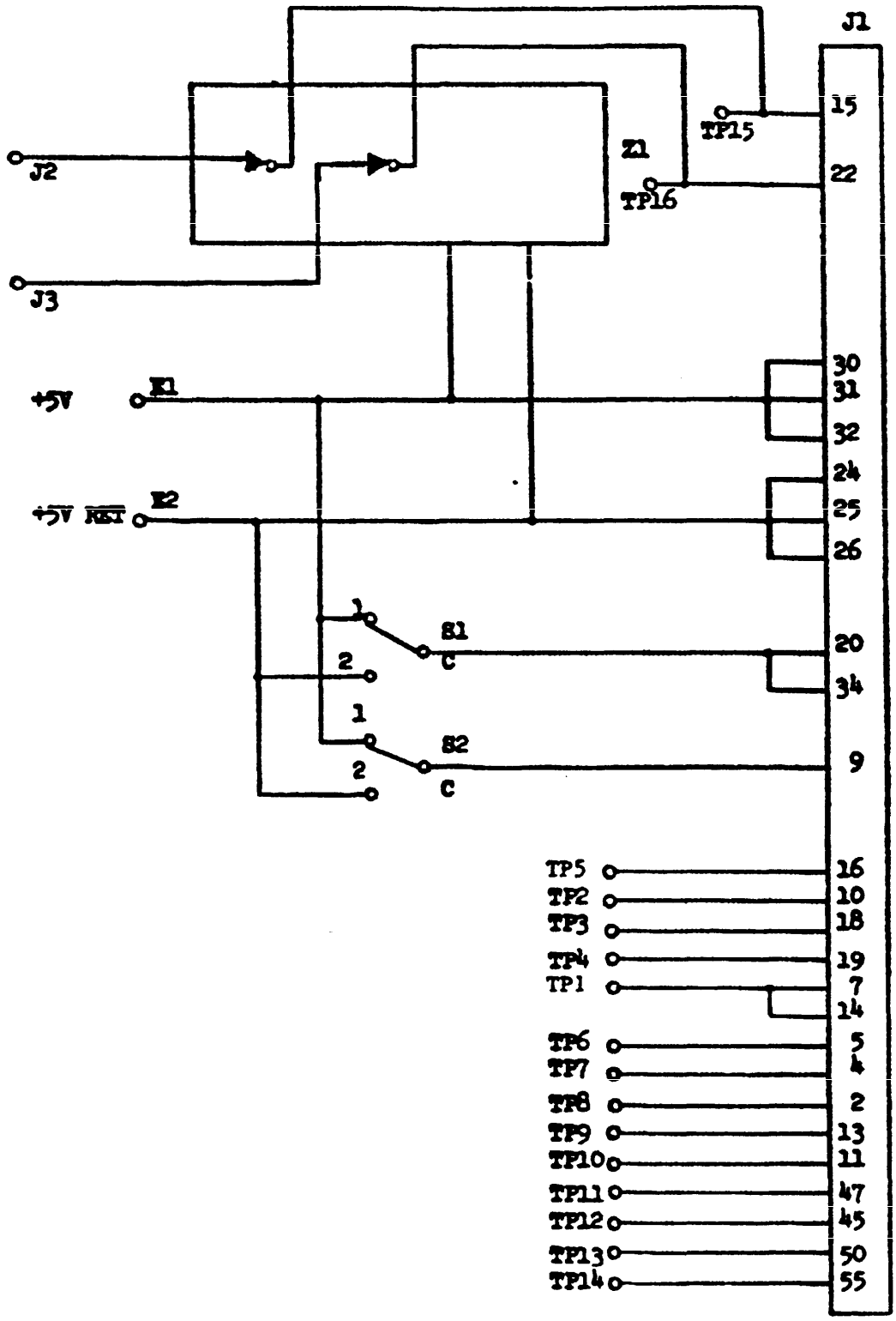


FIGURE 1. TEST FIXTURE SETUP.

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4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. The selected computer timing generators shall be tested as follows: Adjust power supply, Item 1 (see Figure 2), and pulse generators, item 2, to deliver signals as shown in Figure 3, 'Fixture Input Signals". Connect test equipment, items 1 through 4, as shown in Figure 2. Insert computer timing generator into J1 in test fixture, item 4 (Refer Table VI for cross references).

4.6.3.1 T1 and T3 input. Using channels A and B of oscilloscope observe input signals labeled as T1 and T3 on Table 1 on TP16 and TP15 respectively. Synchronize oscilloscope on TP16.

4.6.3.2 T1 and T3 output. Using channels A and B of oscilloscope observe output signals labeled as T1 and T3 on Table II on TP2 and TP5 respectively. synchronize oscilloscope on TP16.

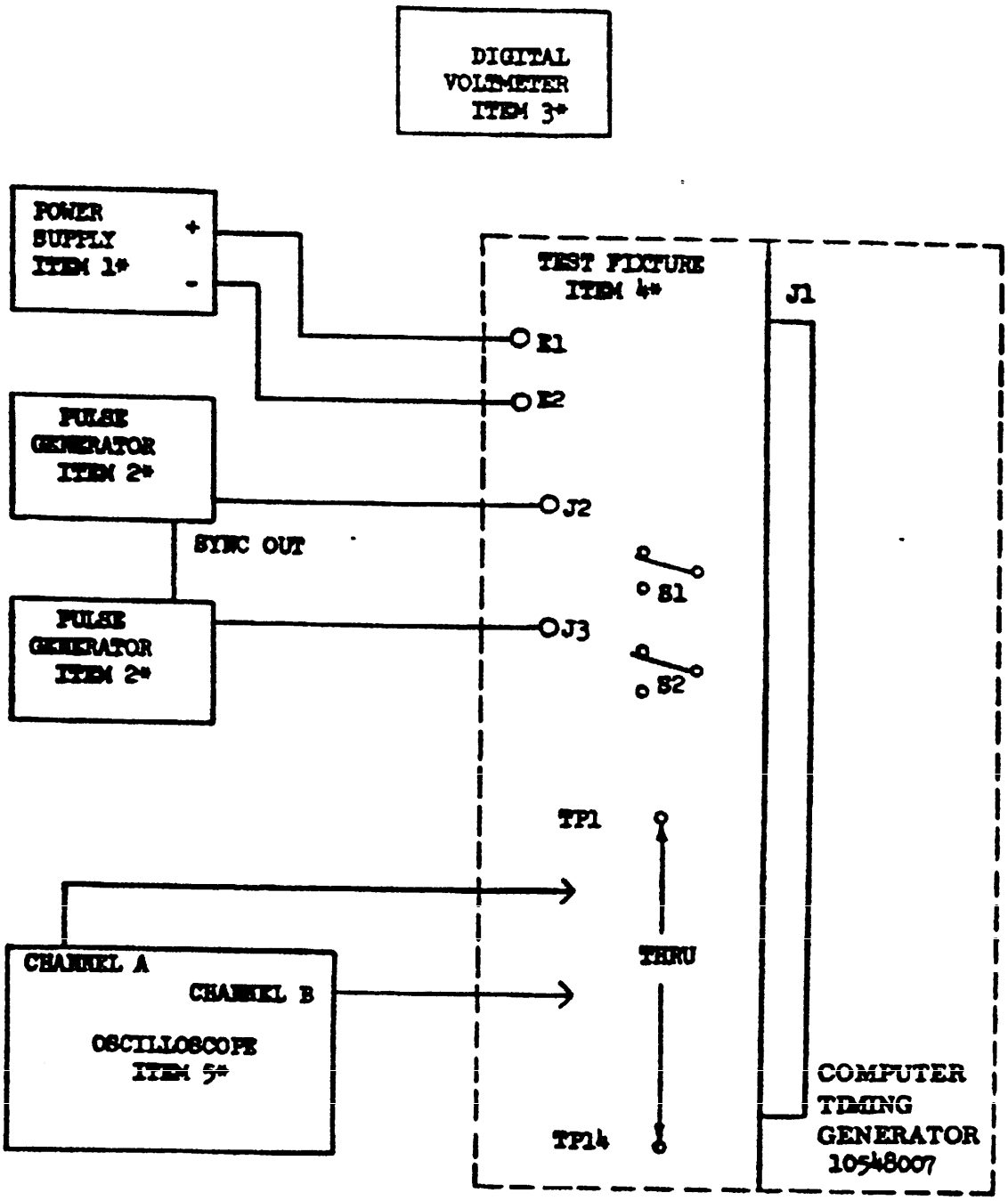
4.6.3.3 T3\4KC output. Connect channel A of oscilloscope to TP4 and channel B to TP5. Synchronize oscilloscope to TP4. observe pulse on channel A with characteristics same as output signal labeled T3 on Table II. There shall also be five pulses on channel B for each pulse on channel A.

4.6.3.4 T1/250 output. Connect channel A of oscilloscope to TP3 and channel B to TP2. Synchronize oscilloscope to TP3. Observe pulse on channel A with characteristics same as input signal labeled T1 on Table II and be coincident with one pulse on channel B. Pulses on channel A shall be spaced every 4 milliseconds (ms).

4.6.3.5 F4/5 and $\overline{F4/5}$ output. Connect channel A of oscilloscope to TP12 and charnel B to TP11. Synchronize oscilloscope on TP4. Observe output signals labeled as F4/5 and $\overline{F4/5}$ on Table III.

4.6.3.6 F2/5 and F2/5 output. Connect channel A of oscilloscope to TP7 and charnel B to TP8. Synchronize oscilloscope , on TP4. observe output signals labeled as F2/5 and $\overline{F2/5}$ on Table III.

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* REFER TO TABLE IV

FIGURE 2. COMPUTER TIMING GENERATOR SETUP.

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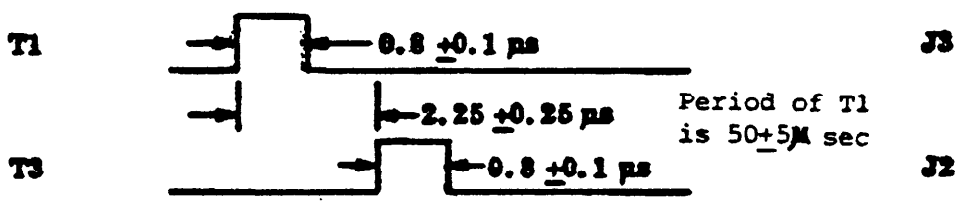


FIGURE 3. FIXTURE INPUT SIGNALS.

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TABLE VI. CROSS REFERENCE FROM SIGNAL TO TEST POINTS.

SIGNAL	BOARD TERMINALS	TEST POINTS
T1	P1-22	TP16
$\overline{\text{T3}}$	P1-15	TP15
T1	P1-10	TP2
T3	P1-16	TP5
T3/4Kc	P1-19	TP4
$\overline{\text{T1/250}}$	P1-18	TP3
F4/5	P1-45	TP12
F4/5	P1-47	TP1l
F2/5	P1-4	TP7
$\overline{\text{F2/5}}$	P1-2	TP8
F1/5	P1-13	TP9
F1/5	P1-11	TP10
200 M	P1-50	TP13
$\overline{\textit{200 M}}$	P1-55	TP14
$\overline{\text{TG SET}}$	P1-7	TP1
$\overline{\text{TG RESET}}$	P1-5	TP6

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4.6 .3.7 ***F1/5 and F1/5*** output. Connect channel A of oscilloscope to TP9 and channel B to TP10. Synchronize oscilloscope on TP4. **observe output signals labeled as F1/5 and F1/5 on Table III .**

4.6.3.8 200 meter counter test. Connect channel A of **oscilloscope** to TP13 and channel B to TP14. Synchronize . **oscilloscope on TP13. with S1 and *S2* in position 1 observe output** signals **labeled as 200 M and M on TP13 and TP14 respectively as** per Table III. When S1 is placed to position 2 **a logic 1 shall be observed on TP13 and a logic 0 on TP14. Return S1 to position 1.**

4.6.3.9 Signal conditioner test. Connect channel A of oscilloscope to TP1 and channel B to TP6. Synchronize OSCilloscope on **TP13. observe output signals labeled as TG SET and TG RESET on TP1 and TP6 respectively as shown on Table III. When S2 is dated to Position 2 a logic 1 shall be observed on TP1 and pulses with the same width and specing as those on T1/250 shall be** observed on TP6.

4.7 Soldering. **As** specified in MIL-C-50652.

5. **PACKAGING**

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet PS10548007. The level of protection shall be as specified in the procurement document.

6. NOTES. **As specified in para 6.0 of MIL-C-50652.**

6.1 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50652/2A	09 April 1993

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY: 10548007 (GENERATOR, COMPUTER TIMING)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Barbara Irwin	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) U.S. Army ARDEC (SMCAR-RAC-S) Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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